

KSHITIZ GOYAL

B.TECH CSE (AI & ML) STUDENT

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PROFILE

Final-year B.Tech CSE (AI & ML) student specializing in building applied AI products and predictive backend systems. Adept at translating complex supply chain and healthcare problems into scalable technical solutions using Python, Deep Learning, and LLM orchestration. Recognized for strong business acumen and product storytelling, placing in the Top 10 out of 600+ teams at the Innovatez GenZ Hackathon 2026.

PROJECTS

SupplyGuard — Applied AI SaaS for Supply Chain Prediction

- Architected an automated risk-scoring engine to predict manufacturing vendor stockouts 14 days prior to assembly line disruptions.
- Engineered a defensive data ingestion pipeline using Pydantic and Pandas to parse unstructured Tally ERP ledgers and isolate corrupted data.
- Integrated a Retrieval-Augmented Generation (RAG) workflow using LangChain and Llama-3 to autonomously synthesize context-aware, factually grounded spot-buy emails to backup suppliers.
- Developed a localized NLP intent classifier to process raw supplier communications (Hinglish/Hindi) and detect delivery delay idioms with high precision.

Bone Fracture Classification Engine

- Developed a multi-class Deep Learning framework to autonomously classify 10 distinct types of bone fractures from raw X-ray imagery.
- Built a custom dataset handler using torch.utils.data and implemented dynamic preprocessing using explicit grayscale conversion and intensity scaling.
- Designed and trained a custom Convolutional Neural Network (CNN) architecture with dropout regularization, achieving 95.9% validation accuracy and 99.3% training accuracy across 20 epochs.

Classical Edge Detection & Segmentation Analyzer — Image Processing

- Engineered an analytical pipeline to benchmark classical computer vision techniques (Canny, Sobel, K-Means, Thresholding) on the BSDS500 dataset.
- Implemented K-Means clustering (K=3) for structural image segmentation, reshaping and mapping spatial data arrays using NumPy.
- Generated automated performance comparison matrices utilizing Seaborn heatmaps to evaluate algorithm efficacy in extracting regional boundaries.

EDUCATION

Bachelor of Technology in CSE (AI & ML)

2023 - 2027

Vellore Institute of Technology (VIT), Bhopal | CGPA: 8.44

PROFESSIONAL SKILLS

- Languages:** Python, SQL, C++ (Basics)
- AI & Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, CNNs, NLP, HuggingFace
- Backend & Frameworks:** FastAPI, Streamlit, LangChain, LlamaIndex
- Databases & Tools:** PostgreSQL, Supabase, Git/GitHub, Docker

ADDITIONAL INFORMATION

- Awards/Activities:** Top 10 Finalist (out of 600+ teams), Innovatez GenZ Hackathon 2026 by Zentiti (developed and pitched SupplyGuard MVP).
- Languages:** English, Hindi, French (Basics).